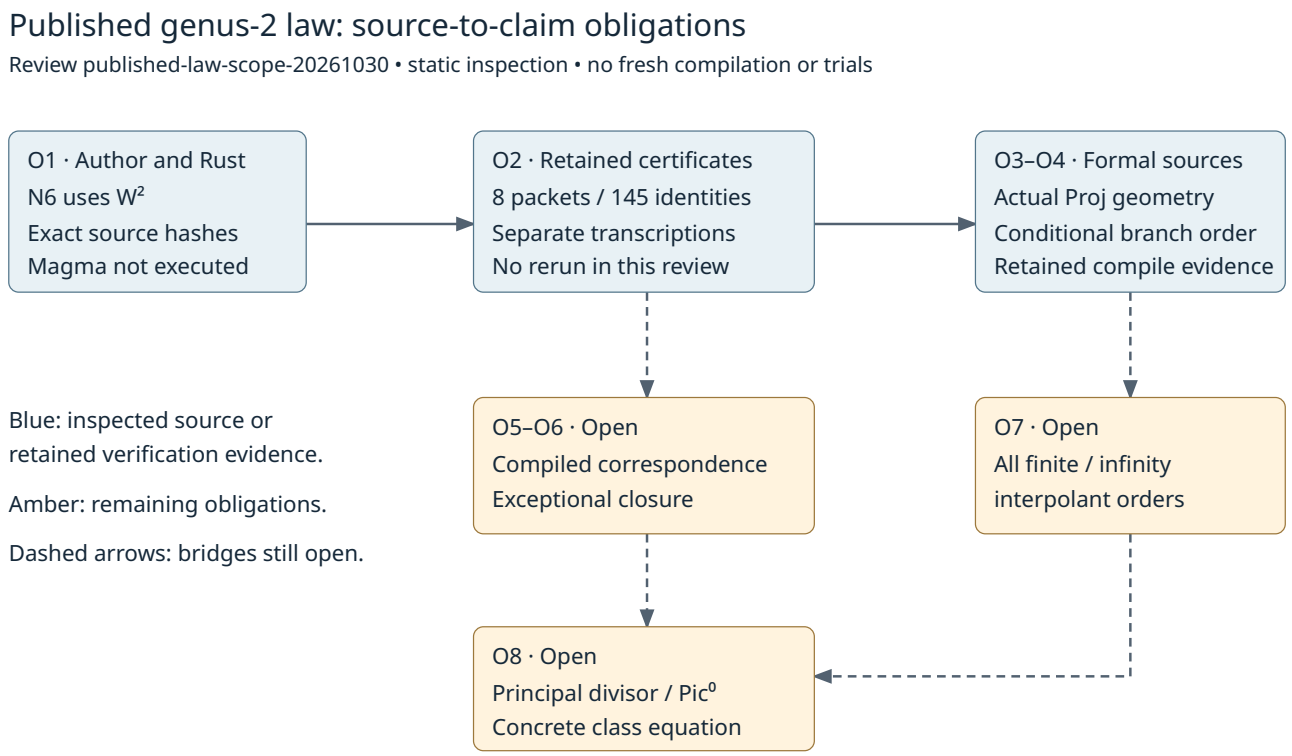
Published genus-2 law: independent scope review

Review ID: published-law-scope-20261030. Task T-930. 30 October 2026.

Scope. Static source-to-claim inspection and retained verification evidence. No fresh compilation, Magma execution, census, solver, replay or benchmark. This packet records obligations and evidence boundaries; it supplies no new cost ratio or universal correctness theorem. The original full research objective remains incomplete.



Reading the diagram. Blue boxes identify inspected source or previously retained evidence, not freshly rerun validation. Amber boxes identify open obligations. Solid arrows organize the evidence chain; they do not assert that a certificate alone proves the next theorem. Dashed arrows mark unresolved proof bridges. Node source paths and statuses are retained in `obligations.json`; the full branch table is in the unchanged `review.md`.

Additive correction, 2026-10-30. Later formal sources already include actual weighted-Proj smoothness, the glued-model-to-Proj isomorphism, and a conditional actual-quintic rational-branch order theorem. Older checklist descriptions that these bridges are wholly missing are stale. Historical files remain unchanged. Compiled semantics and global divisor-class obligations remain open.

Branch obligations and input model

The intended model is a smooth odd-characteristic curve $y^2 = x^5 + a_3x^3 + a_2x^2 + a_1x + a_0$, valid reduced Mumford divisors, nonzero weights and consistent field/curve parameters. Runtime entry points do not check every mathematical hypothesis.

Degree one represents $x = X/Z^2$, $y = Y/(Z^5W)$. Degree two represents $u = x^2 + (Q/Z^2)x + R/Z^4$, $v = (S/(Z^3W))x + T/(Z^5W)$.

Branch	**Observed source behavior**	**Remaining universal obligation**
1+1	Inverse first; then doubling; otherwise distinct-x output.	Nonzero weights, tangency/chord multiplicities and infinity.
Generic 1+2	Nonzero K gives degree two; 16 retained identities.	Runtime-polynomial link, residual sign and principal divisor.
Exceptional 1+2	Cancellation, tripling, or double-plus-other; 12 exception and 16 triple identities.	Exclude mismatch/secondary failure; repeated-root multiplicities.
Same-u 2+2	Inverse, equal-v double, or mixed extracted-point double; mixed 7 identities.	Valid-input zero-slope exclusion and canceled components.
Doubling	B=0 subcases; otherwise C=0 degree drop or degree two; 30 identities.	Closure and valid nonzero output weights on every subcase.
Shared-support	Cancel shared point or double and add remaining points; 27 identities.	None unreachability, including intermediate degree drops.
Coprime drop	B nonzero, C zero gives degree one; 18 identities.	Exhaustive parametrization and residual/infinity multiplicity.
Coprime common	B,C nonzero gives degree two; 19 identities.	Compiled numerator correspondence and all-place divisor equation.

The eight certificate outputs total 145 polynomial identities. They are separately transcribed calculations, not a mechanical proof of every compiled statement. The public projective dispatcher uses `expect`; an unreachable-None proof is a completeness obligation, not a fallback to Cantor arithmetic.

W squared. The exact pinned author `DBL2` defines $WW1 = W1^2$ and uses it in N6. Production Rust also selects W^2. The W^1 variant is a deliberate mutation test, not a demonstrated defect in the published source.

Inversions. Inspected formula branches have no explicit inversion. Nonidentity normalization calls inverse twice (Z and W); identity calls none. This does not establish an inversion-free collector or end-to-end pipeline.

Formal scope, retained context and references

`WeightedProjSmoothFinal.smooth` and `WeightedProjCrossInjective.gluedToProj_isIso` address actual projective geometry. `WeightedProjInfinityDVR` establishes local infinity facts, including order two for Z; it does not yet identify the interpolation function's complete pole divisor.

`WeightedProjSourceBranchActualQuintic` proves order one in an actual projective branch stalk for the proof-side common-2+2 expression under valid input congruences, nonzero determinant, rational branch and support-root hypotheses. Its separable corollary supplies simple-root geometry. It is not a theorem about every compiled Rust branch.

`RuntimeSemantics.residual_represents_input` assumes a principal-divisor/class interface and the required principal-divisor premise. It gives the abstract class consequence; constructing the concrete interface and premise remains necessary. Saved records report Lean checks without `sorryAx`; this review inspected declarations and retained records, without compiling them anew.

Retained finite context, not new measurements. The historical projective-law report records two collector arms, each over 1,260 frozen targets with paired source-rho controls. Its wrapper normalizes every addition; retained state normalizes at relation boundaries. This review does not recompute its counts or prices. Finite successes do not prove universal completeness. The public-source parity report records 34,704 translated-author/Rust comparisons and 34,704 Rust/Cantor comparisons over two finite fields. That was restricted translation of source, not actual Magma execution.

Next proof work. Bind actual runtime expressions and predicates to the existing polynomials; prove exceptional closure and returned-weight nonvanishing; complete interpolant orders at nonbranch points, repeated supports and infinity; construct the global principal divisor and concrete Pic^0 equality. Extend across every branch and handle nonsplit places if the theorem quantifies over arbitrary fields.

References and provenance. Paths are repository-relative; exact inspected file hashes are in `source-hashes.sha256`. The original review is preserved byte-for-byte.

Study root:

`research/projective_law_followup_20261022`.

Relative references below use that root. Full paths and hashes are in the JSON and hash manifest.

- `src/bin/projective_law_gate.rs`; `formal/RuntimeSemantics.lean`.
- `PROOF_AUDIT.md`; `formal/PROJECTIVE_BRANCH_ZERO_ORDER.md`.

  Eight `results/symbolic-*.json` records.
- `REPORT.md`; `public_source_parity/README.md`.

  `n6-homogeneity/source-correction/README.md`.
- Pinned author repository: `https://github.com/ozbayelif/jac-on-jac`.

  Commit `ec98e115ed359c8b7facf559eb0d9f240703f6f5`,

  file `jac-on-jac.m`. Downloaded bytes matched the retained SHA-256; no author program was executed.